Lab Exercise 12 - Start and Access Kubernetes Dashboard

Objective

To enable Kubernetes in Docker Desktop, deploy the Kubernetes Dashboard, and access it securely using a web browser on Windows.

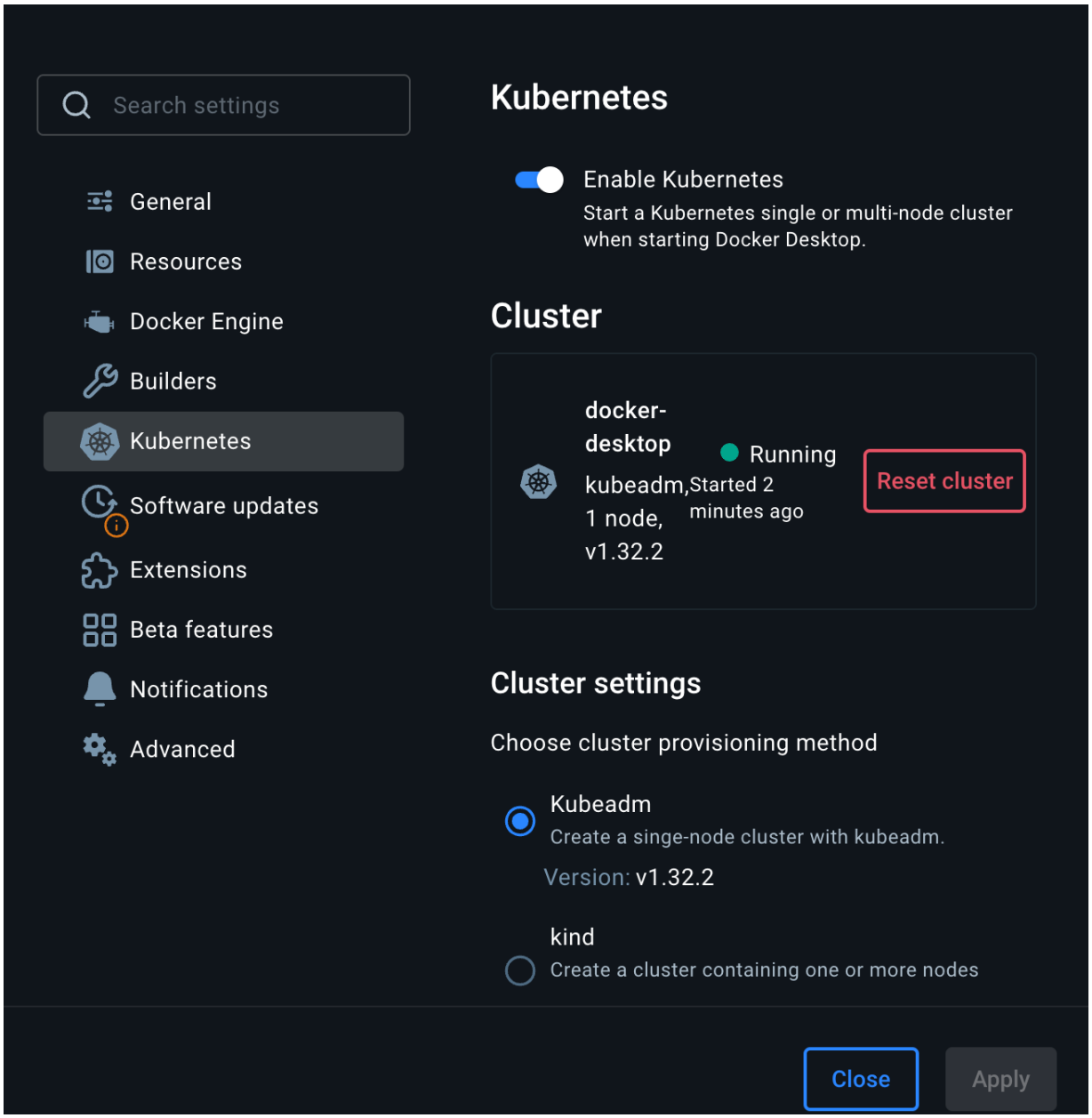
Prerequisites

- Windows 10 / 11
- Docker Desktop installed
- Docker Desktop Kubernetes enabled
- Internet connection
- kubectl (comes bundled with Docker Desktop)

Step 1: Enable Kubernetes in Docker Desktop

1. Open **Docker Desktop**
2. Go to **Settings**
3. Select **Kubernetes**
4. Check **Enable Kubernetes**
5. Click **Apply & Restart**

Wait until Kubernetes status shows **Running** (green).


Search settings
General
Resources
Docker Engine
Builders
Kubernetes
Software updates
Extensions
Beta features
Notifications
Advanced
Kubernetes
Enable Kubernetes
Start a Kubernetes single or multi-node cluster when starting Docker Desktop.
Cluster
docker-desktop
kubeadm, 1 node, v1.32.2
Running
Started 2 minutes ago
Reset cluster
Cluster settings
Choose cluster provisioning method
Kubeadm
Create a singe-node cluster with kubeadm.
Version: v1.32.2
kind
Create a cluster containing one or more nodes
Close
Apply

Step 2: Verify Kubernetes Cluster

Open **PowerShell** or **Command Prompt** and run:

- kubectl version --client
- Check cluster status:
- kubectl cluster-info

Check nodes:

```
kubectl get nodes
```

Expected output:

Node status should be **Ready**

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:63724
CoreDNS is running at https://127.0.0.1:63724/api/v1/namespaces/kube-system/services/kube-dns:dns/prox
y

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
mohdanas@Mohds-MacBook-Air Desktop % kubectl get nodes

NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   78m   v1.35.1
mohdanas@Mohds-MacBook-Air Desktop %
```

Step 3: Deploy Kubernetes Dashboard

Apply the official Kubernetes Dashboard manifest:

```
kubectl apply -f https://raw.githubusercontent.com/kubernetes/dashboard/v2.7.0/aio/deploy/

recommended.yaml
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl apply -f https://raw.githubusercontent.com/kubernetes/das
hboard/v2.7.0/aio/deploy/recommended.yaml

namespace/kubernetes-dashboard configured
serviceaccount/kubernetes-dashboard configured
service/kubernetes-dashboard configured
secret/kubernetes-dashboard-certs configured
secret/kubernetes-dashboard-csrf configured
Warning: resource secrets/kubernetes-dashboard-key-holder is missing the kubectl.kubernetes.io/last-ap
plied-configuration annotation which is required by kubectl apply. kubectl apply should only be used o
n resources created declaratively by either kubectl create --save-config or kubectl apply. The missing
 annotation will be patched automatically.
secret/kubernetes-dashboard-key-holder configured
configmap/kubernetes-dashboard-settings configured
role.rbac.authorization.k8s.io/kubernetes-dashboard configured
clusterrole.rbac.authorization.k8s.io/kubernetes-dashboard configured
rolebinding.rbac.authorization.k8s.io/kubernetes-dashboard configured
deployment.apps/kubernetes-dashboard configured
service/dashboard-metrics-scraper configured
deployment.apps/dashboard-metrics-scraper configured
The ClusterRoleBinding "kubernetes-dashboard" is invalid: roleRef: Invalid value: {"APIGroup":"rbac.au
thorization.k8s.io","Kind":"ClusterRole","Name":"kubernetes-dashboard"}: cannot change roleRef
mohdanas@Mohds-MacBook-Air Desktop %
```

Verify namespace creation:

```
kubectl get ns
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl get ns
NAME                   STATUS   AGE
default                Active   79m
kube-node-lease        Active   79m
kube-public            Active   79m
kube-system            Active   79m
kubernetes-dashboard   Active   78m
```

You should see:

```
kubernetes-dashboard
```

Step 4: Verify Dashboard Pods

Check dashboard pods:

```
kubectl get pods -n kubernetes-dashboard
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl get pods -n kubernetes-dashboard

NAME                                         READY   STATUS    RESTARTS   AGE
dashboard-metrics-scraper-8d46b45f6-bvq7w    1/1     Running   0          100s
kubernetes-dashboard-b44857bbb-lfh29         1/1     Running   0          100s
mohdanas@Mohds-MacBook-Air Desktop %
```

Expected status:

Running

Step 5: Create Admin User for Dashboard Access

Create a service account:

```
kubectl create serviceaccount dashboard-admin  -n kubernetes-dashboard
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl create serviceaccount dashboard-admin  -n kubernetes-dash
board

serviceaccount/dashboard-admin created
mohdanas@Mohds-MacBook-Air Desktop %
```

Create cluster role binding:

```
kubectl create clusterrolebinding dashboard-admin-binding --clusterrole=cluster-admin --

serviceaccount=kubernetes-dashboard:dashboard-admin
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl create clusterrolebinding dashboard-admin-binding --clust
errole=cluster-admin   --serviceaccount=kubernetes-dashboard:dashboard-admin

clusterrolebinding.rbac.authorization.k8s.io/dashboard-admin-binding created
mohdanas@Mohds-MacBook-Air Desktop %
```

Step 6: Generate Dashboard Login Token

Run the following command to get the token:

```
kubectl -n kubernetes-dashboard create token dashboard-admin
```

eyJhbGciOiJSUzI1NiIsImtpZCI6Imp5OGxKMkRZYzd5UzVzVzU1RVBoQktrQVZPVktqcnp0WGJxTX
hvWlUwQ00ifQ.eyJhdWQiOlsiaHR0cHM6Ly9rdWJlcm5ldGVzLmRlZmF1bHQuc3ZjLmNsdXN0ZXIu
bG9jYWwiXSwiZXhwIjoxNzcyMjIxMTU2LCJpYXQiOjE3NzIyMTc1NTYsImlzcyI6Imh0dHBzOi8va3
ViZXJuZXRlcy5kZWZhdWx0LnN2Yy5jbHVzdGVyLmxvY2FsIiwianRpIjoiNWE1MTc0NjctM2ZkOC0
0Nzc4LTk1NmYtMzhiY2E0NjdmY2VkIiwia3ViZXJuZXRlcy5pbyI6eyJuYW1lc3BhY2UiOiJrdWJlcm5
ldGVzLWRhc2hib2FyZCIsInNlcnZpY2VhY2NvdW50Ijp7Im5hbWUiOiJkYXNoYm9hcmQtYWRtaW4i
LCJ1aWQiOiJkMWZiZjkwNy04OTc4LTRkNDEtYWU2Yy00YmU1MDg5MDllOTYifX0sIm5iZiI6MT
c3MjIxNzU1Niwic3ViIjoic3lzdGVtOnNlcnZpY2VhY2NvdW50Omt1YmVybmV0ZXMtZGFzaGJvYXJ
kOmRhc2hib2FyZC1hZG1pbiJ9.220MIRQ4PReJZVQuKJqEfXwXuaJiqqY7jty_kJlqqa3npXjmKz2WjH
Nyimy2Qgfc5J0us7Bx-U4BU0AEZaIpXDsEGdOnmVbU0HUT1gdtGrkeBkXE0KOgbZf-
fi1QjDN155e069G9Ng_mBoJICEuzUD_s09lem8K-
EcyfQYMs5BxdicIrOJQgVYxRQI7JIOVcDJXZ2cYUDKTHYRH-
ZnKXfW6dOH4cUtk5PGnJ3xaz2RaHdQVNu5EteSGttqEd6D6epnMtMQJYZTsQkN8rRY5fjm_eF9rz_0
vIrGqd3A8Yie8Nsc8H_m5nPse-UbGR2gRLVjD7-SZYvv3Fmq1Rs-r5qg

Copy the generated token (you will paste it in the browser later).

Step 7: Start Kubernetes Dashboard

Run the proxy command:

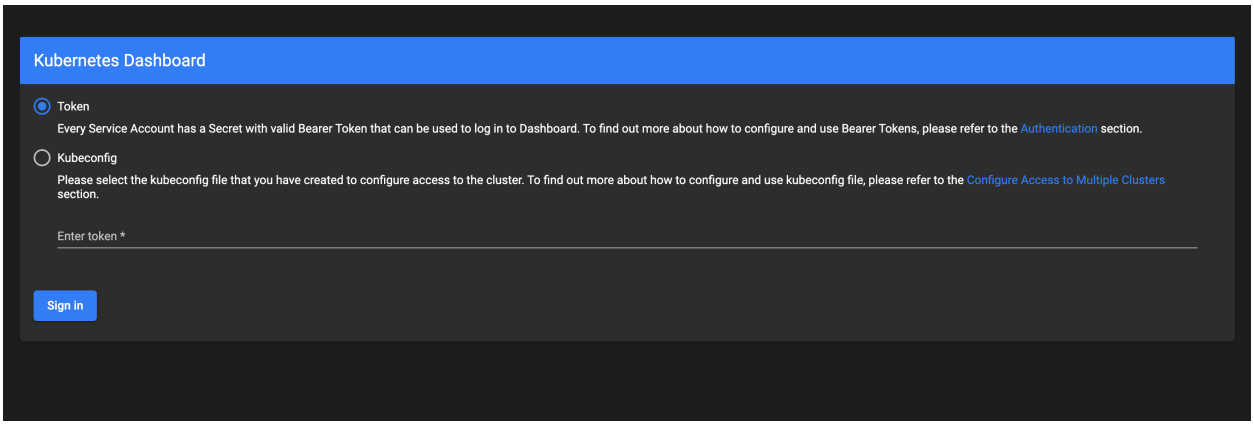
```
kubectl proxy
```

Keep this terminal **running**.

Step 8: Access Kubernetes Dashboard in Browser

Open a web browser and paste the following URL:

```
http://localhost:8001/api/v1/namespaces/kubernetes-dashboard/services/https:kubernetes-dashboard:/proxy/
```



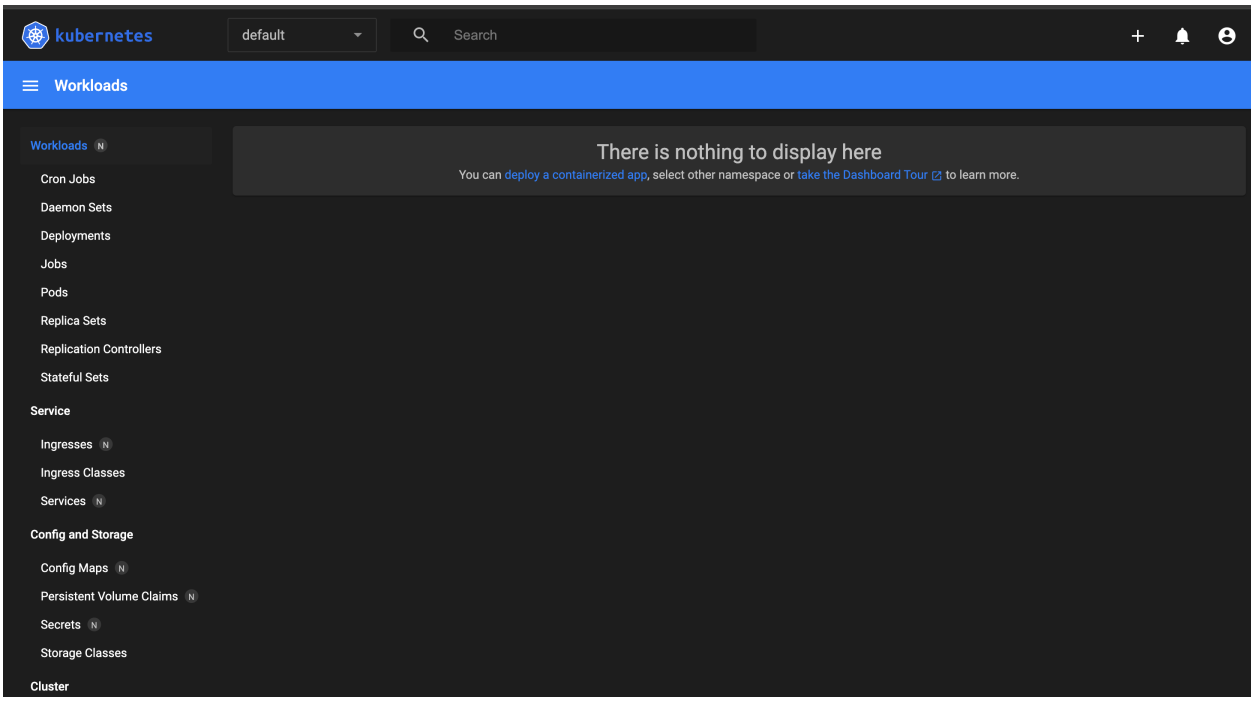
Step 9: Login to Dashboard

1. Select **Token** authentication
2. Paste the token generated earlier
3. Click **Sign In**

You should now see the **Kubernetes Dashboard UI**.

Step 10: Explore Dashboard

You can now view:

- Nodes
- Pods
- Deployments
- Services
- Namespaces

- ConfigMaps and Secrets